

Exercise

- Consider the following requirement for a software to be developed for controlling a chemical plant. The chemical plant has a number of emergency conditions. When any of the emergency conditions occurs, some prespecified actions should be taken. The different emergency conditions and the corresponding actions that need to be taken are as follows.
 - If the temperature of the chemical plant exceeds T1 degs, then the water shower should be turned ON and the heater should be turned OFF.
 - If the temperature of the chemical tank falls below T2 degs, then the heater should be turned ON and the water shower should be turned OFF.
 - If the pressure of the chemical plant is above P1, then the valve V1 should be OPENED.
 - If the chemical concentration of the tank rises above M, and then temperature of the tank is more than T3 degs, then the water shower should be turned ON.
 - If the pressure rises above P3 and the temperature rises above T1 degs, then the water shower should be turned ON, valves V1 and V2 are OPENED and the alarm bells sounded.

Do the following

- Write the requirements of this chemical plant software in the form of decision table.
- Also draw a decision tree to represent the above processing logic.

Condition	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Temperature > T1	TRUE				TRUE
Temperature < T2		TRUE			
Pressure > P1			TRUE		
Pressure > P3					TRUE
Concentration > M				TRUE	
Temperature > T3				TRUE	
Action					
Water Shower Turned ON	Yes / √			Yes / √	Yes / √
Water Shower Turned OFF		Yes / √			
Heater Turned OFF	Yes / √				
Heater Turned ON		Yes / √			
OPEN Valve V1			Yes / √		Yes / √
OPEN Valve V2					Yes / √
Sound Alarm Bell					Yes / √

Rows = 13

Columns = 5

Condition	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Temperature	> T1	< T2		> T3	> T1
Pressure			> P1		> P3
Concentration				> M	
Action					
Water Shower	ON	OFF		ON	ON
Heater	OFF	ON			
Valve V1			OPEN		OPEN
Valve V2					OPEN
Alarm Bell					SOUND

Rows = 8

Columns = 5

Exercise

The machine waits for the start switch to be pressed. After the user presses the start switch, the machine fills the wash tub with either hot or cold water depending upon the setting of the Hot-Wash switch. The water filling continues until the high level is sensed. The machine starts the agitation motor and continues agitating the wash tub until either the preset timer expires or the user presses the stop switch. After the agitation stops, the machine waits for the user to press the Start-Drying switch. After the user presses the Start-Drying switch, the machine starts the hot air blower and continues blowing hot air into the drying chamber until either the user presses the stop switch or the preset timer expires.

Represent the decision making involved in the operation of the above washing machine by means of decision table.

Condition	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
Press START Switch	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
Hot-Wash-Switch OFF	TRUE					
Hot-Wash-Switch ON		TRUE				
Preset Timer Expired			FALSE			FALSE
Press STOP Switch				TRUE	FALSE	
Press Start-Drying-Switch					TRUE	TRUE
Water Level (= HIGH)	FALSE	FALSE				
Action						
Fill Wash Tub with COLD water	Yes / √					
Fill Wash Tub with HOT water		Yes / √				
Start Agitation Motor			Yes / √			
Stop Agitation Motor				Yes / √		
Start Hot Air Blower					Yes / √	Yes / √

Exercise – Decision Trees

SafeHome software enables the homeowner to configure the security system when it is installed, monitors all sensors connected to the security system, and interacts with the homeowner through a keypad and function keys contained in the SafeHome control panel. During installation, the SafeHome control panel is used to program and configure the system. Each sensor is assigned a number and type, a master password is programmed for arming and disarming the system, and telephone numbers are input for dialing when a sensor event occurs. When a sensor event is recognized, the software invokes an audible alarm attached to the system. After a delay time that is specified by the homeowner during system configuration activities, the software dials a telephone number of a monitoring service, provides information about the location, reporting and the nature of the event that has been detected. The telephone number will be redialed every 20 seconds until telephone connection is obtained. All interaction with SafeHome system is managed by a user-interaction subsystem that reads input provided through the keypad and function keys, displays prompting messages and system status on the LCD display.

Represent processing logic for SafeHome system using decision tree and table.

Exercise

Identify functional requirements based on the following description for the Medical Shop Automation (MSA) software that should help the shop owner save maximum time and should be prioritized. Each of the requirement specification should at minimum contain a requirement identification number, short description (few words), inputs, outputs, pre-requisites, and necessary processing details for it.

A retail medicine shop deals with a large number of medicines procured from various manufacturers. The shop owner maintains different medicines in wall mounted and numbered racks. The shop owner maintains as few inventory for each item as reasonable, to reduce inventory overheads after being inspired by the just-in-time (JIT) philosophy. Thus, one important problem the shop owner faces is to be able to order items as soon as the number of items in the inventory reduces below a threshold value. The shop owner wants to maintain medicines to be able to sustain selling for about one week. To calculate the threshold value for each item, the software must be able to calculate the average number of medicines sales for one week for each part. At the end of each day, the shop owner would request the computer to generate the items to be ordered. The computer should print out the medicine description, the quantity required, and the address of the vendor supplying the medicine. The shop owner should be able to store the name, address, and the code numbers of the medicines that each vendor deals with. Whenever new supply arrives, the shop owner would enter the item code number, quantity, batch number, expiry date, and the vendor number. The software should print out a cheque favoring the vendor for the items supplied. When the shop owner procures new medicines it had not dealt with earlier, he should be able to enter the details of the medicine such as the medicine trade name, generic name, vendors who can supply this medicine, unit selling and purchasing price. The computer should generate a code number for this medicine which the shop owner would paste as the code number in the rack where this medicine would be stored. The shop owner should be able to query about a medicine either using its generic name or the trade name and the software should display its code number and the quantity present. At the end of every day the shop owner would give a command to generate the list of medicines which have expired. It should also prepare a vendor-wise list of the expired items so that the shop owner can ask the vendor to replace these items. Currently, this activity alone takes a tremendous amount of labour on the part of the shop owner and is a major motivator for the automation endeavor. Whenever any sales occurs, the shop owner would enter the code number of each medicine and the corresponding quantity sold. The MSA should print out the cash receipt. The computer should also generate the revenue and profit for any given period. It should also show vendor-wise payments for the period.

Prepare functional requirements based on the above system description. Grade your own document against all the characteristics of a good requirements document. In case, you find that you have compromised the quality of your document, give suitable reasons for doing so. Represent processing logic for the above system description using decision trees/tables.